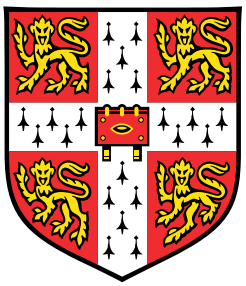
Harnessing Transformations in Quantum Computing

Fermion-to-Qubit Mappings, Quantum Metropolis Sampling and Trapped-Ion Experiments



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Doctor of Philosophy

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I would like to dedicate this thesis to my loving parents ...

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Oliver O'Brien
November 2026

Acknowledgements

And I would like to acknowledge ...

Abstract

This is where you write your abstract ...

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Nomenclature

Roman Symbols

F	complex function

Greek Symbols

γ	a simply closed curve on a complex plane
$\imath$	unit imaginary number $\sqrt{-1}$
π	$\simeq 3.14\ldots$

Superscripts

j	superscript index

Subscripts

0	subscript index
crit	Critical state

Other Symbols

$\oint_\gamma$	integration around a curve γ

Acronyms / Abbreviations

ALU	Arithmetic Logic Unit
BEM	Boundary Element Method
CD	Contact Dynamics
CFD	Computational Fluid Dynamics
CIF	Cauchy's Integral Formula
CK	Carman - Kozeny

DEM Discrete Element Method

DKT Draft Kiss Tumble

DNS Direct Numerical Simulation

EFG Element-Free Galerkin

FEM Finite Element Method

FLOP Floating Point Operations

FPU Floating Point Unit

FVM Finite Volume Method

GPU Graphics Processing Unit

LBM Lattice Boltzmann Method

LES Large Eddy Simulation

MPM Material Point Method

MRT Multi-Relaxation Time

PCI Peripheral Component Interconnect

PFEM Particle Finite Element Method

PIC Particle-in-cell

PPC Particles per cell

RVE Representative Elemental Volume

SH Savage Hutter

SM Streaming Multiprocessors

USF Update Stress First

USL Update Stress Last

Chapter 1

Introduction

1.0.1 Hybrid Mappings

Simulating quantum systems is one of the most promising applications of quantum computing in the near future [25, 41, 52]. The class of fermionic systems is particularly important and simulating them has an impressive range of applications: from problems in quantum chemistry [47, 55], to the Hubbard model in condensed matter physics [1], to Lattice Gauge Theories [5, 12] and beyond. Classical methods applied to these problems do not scale favourably with the system size, whereas quantum techniques (e.g. VQE [48], Trotterization [64] and others [8, 27, 53]) open up efficient avenues for studying static and dynamic properties of fermionic systems. In recent years, there has been a considerable experimental effort to simulate small fermionic systems on a quantum computer [6, 40].

One of the crucial challenges facing these simulations is encoding the anti-commuting fermionic variables as operators acting on the qubit degrees of freedom. A desirable transformation would map interactions in the fermion picture to qubits resulting in (quasi-) local operators for the latter side. Furthermore, it should be engineered in a way to substantially reduce gate overhead in a circuit model – leading to faster simulation times and more robustness with respect to errors.

There is currently no known fermion-to-qubit mapping that preserves perfect locality and consumes no auxiliary resources (in the form of qubits or the extra gate overhead) for an arbitrary interaction graph. However, in the last decade there has been remarkable progress towards designing efficient mappings [66, 58, 32, 8, 56, 21, 16, 61] and exploring new ideas that would improve the existing ones [18]. A summary of select mappings (including the most efficient ones) can be found in Table 2.1. These mappings fall into two main categories: non-local mappings which require no additional resources but produce gate counts dependant on the total number of fermionic modes; local mappings which use additional resources (ancilla qubits) to produce constant gate counts independent of the number of fermionic modes. These local mappings can be physically interpreted as lattice gauge theories [15]. The best-known "explicitly constructed" local mapping is due to Chen and Xu [16]. It uses the fewest qubits per fermion mode (1.25) while producing Hamiltonians with k-local average and maximum Pauli string, for constant k. There do exist general constructions with lower ratios that only require $1+\frac{1}{k}$ qubits per fermion however they have the trade-off of a Pauli weight growing with $O(k)$.

In this work, we introduce a family of parametrized Hybrid mappings (Section 2.1.1) that combine the relaxed connectivity constraints of the Jordan-Wigner mapping and the increased locality of the Bravyi-Kitaev mapping to produce drastically reduced gate counts [63]. In the context of limited qubit connectivity (Section 2.1.2) the Hybrid mapping produces gate counts that scale with $\frac{N}{2n}$ compared to $\frac{N}{2}$ for the Jordan-Wigner and Bravyi-Kitaev mappings on an $N \times N$ lattice (where $n \ll N$). In the regime of all-to-all connectivity the Hybrid mapping gate count scales with $\frac{N}{2n^2}$ (Section 2.1.2) even further outperforming the unaffected Jordan-Wigner mapping. However, in the absence of connectivity constraints the Bravyi-Kitaev mapping gate count scales with $O(\log N)$. Despite this our numerical analysis in Section 2.1.2 shows the Hybrid mapping still results in lower gate counts than the Bravyi-Kitaev mapping for small lattices (e.g. lattices smaller than 164×164 for $n = 4$).

We also construct a Hybrid+ family of parametrized local mappings (Section 2.1.1) which only require $1 + \frac{1}{n^2}$ qubits per fermionic mode. Our construction interpolates between the Auxillary Qubit mapping [66, 61] and the Bravyi-Kitaev mapping by setting $n = 1$ for the former and $n = N$ for the latter. For all the intermediate n, our construction results in novel highly efficient mappings that produce a dramatic reduction in the number of auxiliary qubits required, while simultaneously generating local Pauli strings. We illustrate examples for $n = 4$ and $n = 8$ which demonstrate 1.0625 and 1.0156 qubit-to-fermion ratios compared to the best existing local mapping [16] which achieves a 1.25 qubit-to-fermion ratio.

The key insight of our work is that it is possible to apply seemly incompatible mappings on different scales such that their relative strengths complement each other. We combine the efficiency of the Bravyi-Kitaev mapping with the low connectivity restraints of the Jordan-Wigner mapping to produce a far superior Hybrid mapping. On a small scale (within each cell) we use the Bravyi-Kitaev mapping (in few fermion regime when it's connectivity restraints are not too large), but when we zoom out to a larger scale we use the Jordan-Wigner mapping between the cells. By varying the scale at which each mapping is dominant we are able to find an optimal mapping for a given quantum computer.

1.1 Preliminaries

In this section, we introduce the mappings our Hybrid schemes extend: the Jordan-Wigner Mapping (Section 1.1.1), the Bravyi-Kitaev Mapping (Section 1.1.2), and the Auxiliary Qubit Mapping (Section 1.1.3).

1.1.1 Jordan-Wigner Mapping

The Jordan-Wigner transform [39] is arguably the oldest known fermion-to-qubit mapping. It stores the occupation number of the i-th fermionic mode in the i-th qubit. Therefore, all the non-local behaviour (the

parity information) has to be encoded in the operators [18]:

$$a_i \to \left(\bigotimes_{k=1}^{i-1} Z_k\right) \sigma_i^-$$
$$a_i^\dagger \to \left(\bigotimes_{k=1}^{i-1} Z_k\right) \sigma_i^+$$

where

$$\sigma_i^- = \frac{1}{2}(X_i + iY_i) = |0\rangle_i \langle 1|_i$$
$$\sigma_i^+ = \frac{1}{2}(X_i - iY_i) = |1\rangle_i \langle 0|_i$$

The $\bigotimes_{k=1}^{i-1} Z_k$ term is necessary to apply the parity phase shift $(-1)^{\sum_{k=0}^{i-1} f_k}$, where f_k are defined below. These operators act in exactly the same way on a qubit spin basis as the fermionic creation and annihilation operators act upon the Fock basis in Eqns (1.1) and (1.2).

$$a_p |f_0, \cdots, f_i, \cdots, f_{M-1}\rangle = \delta_{f_p,1}(-1)^{\sum_{i=0}^{p-1} f_i} |f_0, \cdots, f_p \oplus 1, \cdots, f_{M-1}\rangle \tag{1.1}$$

$$a_p^\dagger |f_0, \cdots, f_i, \cdots, f_{M-1}\rangle = \delta_{f_p,0}(-1)^{\sum_{i=0}^{p-1} f_i} |f_0, \cdots, f_p \oplus 1, \cdots, f_{M-1}\rangle \tag{1.2}$$

1.1.2 Bravyi-Kitaev Mapping

Our construction may be regarded as a particular extension of the Bravyi-Kitaev mapping [9]. Previous extensions of the Bravyi-Kitaev mapping have combined the Bravyi-Kitaev fast and super-fast mappings to logarithmically reduce the Pauli weight of fermionic operators [59].

The Bravyi-Kitaev mapping stores the parity of fermionic modes in a non-local Fenwick tree data structure [32]. A Fenwick tree [24] is a partial ordering of binary representations where every node is a copy of its parent with one of the 1s changed to a 0 as shown in Figure 1.1. In this section we outline how the Bravyi-Kitaev mapping encodes m fermionic modes:

First, each fermionic mode is assigned a corresponding qubit and a node in a Fenwick tree. Every qubit stores the total parity of the fermionic modes below it in the Fenwick tree. This allows fermionic operators to be constructed with Pauli-weight $O(\log m)$.

These operators (as shown in [56]) are constructed using three easily calculated subsets of the Fenwick tree: the Update (U), Parity (P), and Flip (F) sets.

$$\begin{aligned} a_i &\to X_{U_i} \sigma_i^- Z_{P_i/F_i} \\ &\to \frac{1}{2}\left(\bigotimes_{j\in U_i} X_j \otimes X_i \otimes \bigotimes_{j\in P_i} Z_j + i \bigotimes_{j\in U_i} X_j \otimes Y_i \otimes \bigotimes_{j\in P_i/F_i} Z_j\right) \end{aligned}$$

$$a_i^\dagger \to X_{U_i}\sigma_i^+ Z_{P_i/F_i}$$
$$\to \frac{1}{2}\left(\bigotimes_{j\in U_i} X_j \otimes X_i \otimes \bigotimes_{j\in P_i} Z_j - i\bigotimes_{j\in U_i} X_j \otimes Y_i \otimes \bigotimes_{j\in P_i/F_i} Z_j\right)$$

Each of the above sets is defined (similarly to the original formulation [56]) as follows :

Use the binary representation: $\alpha = \alpha_{l-1}\ldots\alpha_0$ and $\beta = \beta_{l-1}\ldots\beta_0$, where index 0 denotes the most significant bit. Define $\prec$ to represent the partial ordering of the Fenwick tree. Fermionic modes are zero-indexed.

Definition 1. *Let the Update Set $U(\alpha)$ be such that $\beta \in U(\alpha)$ if and only if for some i_0, $\alpha_{i_0} = 0$ and $\alpha_i = \beta_i$ for $i \geq i_0$ and $\beta_i = 1$ for $i \leq i_0$. Equivalent to $\beta \in U(\alpha)$ if and only if $\alpha \prec \beta$.*

Definition 2. *Let the Parity set $P(\alpha)$ contain elements $\beta \in P(\alpha)$ if and only if for some i_0: $\alpha_{i_0} = 1$, $\beta_{i_0} = 0$, $\beta_i = \alpha_i$ for $i > i_0$ and $\beta_i = 1$ for $i < i_0$, then $\sum_{j\in P(i)} q_j = \sum_{j<i} f_j$*

Definition 3. *Let the Flip set $F(\alpha)$ contain elements $\beta \in F(\alpha)$ if and only if for some i_0: $\beta_{i_0} = 0$, $\beta_i = \alpha_i$ for $i \neq i_0$ and $\alpha_i = 1$ for $i \leq i_0$, then $\sum_{j\in F(i)} q_j = \sum_{j\prec i} f_j$*

Examples of these sets are given in Figure 1.1. We present three simple algorithms for generating these sets which will later be used in our construction: GENERATEUPDATESET, GENERATEPARITYSET, and GENERATEFLIPSET.

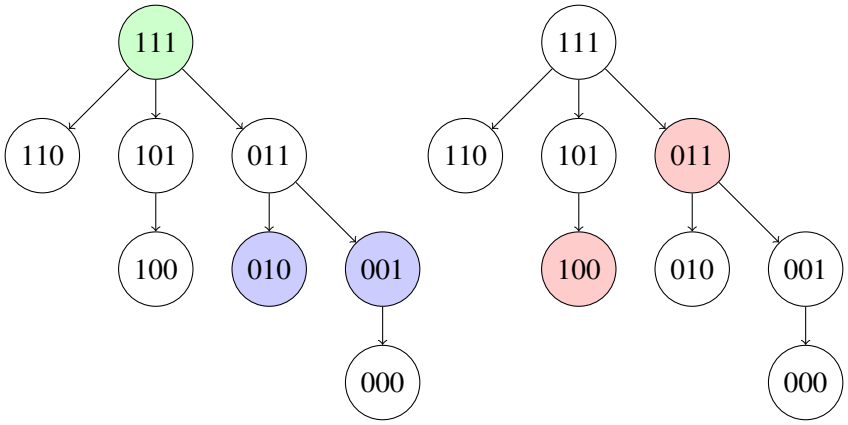


Fig. 1.1 **Top:** Update set (green) and Flip set (blue) for $i = 3$ (011) on Fenwick tree with $N = 8$. **Bottom:** Parity set (red) for $i = 5$ (101) on Fenwick tree with $N = 8$.

1.1.3 Auxiliary Qubit Mapping

Though originally pioneered by [66], here we recall the formalization of the auxiliary mappings from [61]. The principle behind auxiliary mapping schemes is to create stabilisers which are similar to non-local interaction terms in the Hamiltonian, so they can be used to cancel each other out. For example, consider an n-qubit Hamiltonian consisting of m Pauli-strings h_i some of which contain a non-local Pauli-string p:

$$H = \sum_{i=0}^{m} h_i$$

Algorithm 1 GENERATEUPDATESET

Input: Fermionic enumeration: α, Number of fermionic modes: n
Output: Update Set U_α

updateSet = { }
$l \leftarrow$ length of binary representation of α
$\beta \leftarrow \alpha$
for i counting from 0 to $l-1$ **do**
    if α_i is 0 **then**
        $\beta \leftarrow \beta_{l-1} \ldots \beta_{i+1} 1 \beta_{i-1} \ldots \beta_0$
        if $\beta < n$ **then**
            Add β to updateSet
        end if
    end if
end for

Algorithm 2 GENERATEPARITYSET

Input: Fermionic enumeration: α
Output: Parity Set P_α

paritySet = { }
$l \leftarrow$ length of binary representation of α
for i counting down from $l-1$ to 0 **do**
    if α_i is 1 **then**
        $\beta \leftarrow \alpha_{l-1} \alpha_{l-2} \ldots \alpha_{i+1} 01 \ldots 1$
        Add β to paritySet
    end if
end for

Algorithm 3 GENERATEFLIPSET

Input: Fermionic enumeration: α
Output: Flip Set F_α

flipSet = { }
$l \leftarrow$ length of binary representation of α
$i \leftarrow 0$
while α_i is 1 **do**
    $\beta \leftarrow \alpha_{l-1} \ldots \alpha_{i+1} 0 \alpha_{i-1} \ldots \alpha_0$
    Add β to flipSet
    $i \leftarrow i+1$
end while

- Let $|\psi\rangle_S$ be an arbitrary state on the n-qubit system ($\mathscr{H}_S$)
- Introduce an ancillary qubit $|0\rangle_A$ ($\mathscr{H}_A$)
- Entangle the ancilla with the n-qubit state by applying a given unitary V

$$V|\psi\rangle_S|0\rangle_A = |\tilde{\psi}\rangle_{SA}$$

- Find σ such that $(p \otimes \sigma)$ is a stabiliser:

$$(p \otimes \sigma)|\tilde{\psi}\rangle_{SA} = |\tilde{\psi}\rangle_{SA}$$

- Modify H so it commutes with the stabiliser (1.3) and thus acting with the modified Hamiltonian $\tilde{H}$ on a stabiliser state always produces another stabiliser state. This involves adding an operator κ to each term h_i in the Hamiltonian that anti-commutes with $(p \otimes \sigma)$ to produce a commuting term $h_i^{(\kappa)} = h_i \otimes \kappa$. This modification must be such that its action on the entangled system is equivalent to the action of the original Hamiltonian on the n-qubit state (1.4).

$$H^{(\kappa)}(p \otimes \sigma)|\tilde{\psi}\rangle_{SA} = (p \otimes \sigma)H^{(\kappa)}|\tilde{\psi}\rangle_{SA} \tag{1.3}$$

$$H^{(\kappa)}|\tilde{\psi}\rangle_{SA} = V(H \otimes I)|\psi\rangle_S|0\rangle_A \tag{1.4}$$

- Now we can use the stabiliser to cancel out the non-local p string in the Hamiltonian to get a simplified Hamiltonian $\tilde{H}$ which is logically identical.

$$\tilde{h}_i = \begin{cases} h_i^{(\kappa)}(p \otimes \sigma) & p \in h_i \\ h_i^{(\kappa)} & p \notin h_i \end{cases}$$

$$\tilde{H} = \sum_{i=0}^{m} \tilde{h}_i$$

To perform this procedure to remove multiple non-local strings p_i, we can simply repeat the process with additional ancillas. However as the stabilisers $p_i \otimes \sigma_i$ must share common stabiliser states, it is only possible to simultaneously remove commuting strings.

Chapter 2

Fermion-to-Qubit Mappings

2.1 Hybrid Fermion-to-Qubit Mappings

We present two novel mapping algorithms: Hybrid mapping (Section 2.1.1) and Hybrid+ mapping (Section 2.1.1). The Hybrid mapping combines the low connectivity constraints of the Jordan-Wigner mapping with the increased locality of the Bravyi-Kitaev mapping. The Hybrid+ mapping builds on the strengths of the Hybrid mapping and expands the mapping to include aspects of AQM [66, 61] to increase the locality using ancilla qubits. In this section we outline their construction in detail, and then in Section 2.1.2 we will analyse their performance. Both these mappings have been designed for a square-lattice fermionic interaction graph. For the sake of clarity, they are both described with $n \times n$ cells, however, it is trivial to extend them to rectangular $l \times h$ cells. It is also possible to perform these mappings with irregular varying cell sizes. We note that it is possible to extend this mapping to any connectivity or fermionic interaction graph, one must simply make an appropriate choice of cells to split the graph into and then pick an order in which to connect them via Jordan-Wigner.

2.1.1 Hybrid mapping algorithm

Input: *$N \times N$ Fermion Square-Lattice Interaction Graph, Cell Shape $n \times n$, $1 \leq n \leq N$* and n is a factor of N
Output: *Pauli-strings corresponding to interaction edges, Enumerated square-lattice of qubits*
Summary: This mapping effectively splits up the square lattice into a series of Fenwick trees (Bravyi-Kitaev maps) and connects them via a Jordan-Wigner mapping on the roots, as demonstrated in Figures 2.1 to 2.5. These diagrams show how creation and annihilation operators are mapped to Jordan-Wigner like $ZZ \ldots ZX$ strings on the roots of previous trees and Bravyi-Kitaev like strings within the current tree.
Algorithm:

1. Divide the square lattice of fermions into cells of size $n \times n$

	Qubit to fermion ratio	Max Pauli string	Avg. Pauli string	Avg. Range	Max Range
Jordan-Wigner [39]	1	$2N$	$\frac{N}{2}+\frac{3}{2}$	$\frac{N}{2}+\frac{3}{2}$	N+1
Bravyi-Kitaev [8]	1	$\mathscr{BK}_{N^2}$	$\overline{\mathscr{BK}}_{N^2}$	$O(\frac{N}{2})$*	O(N)
$n\times n$ **Hybrid**	1	$\frac{2N}{n}+$ $2\mathscr{BK}_{n^2}$	$\frac{N}{2n^2}+$ $\frac{n+1}{n}\overline{\mathscr{BK}}_{n^2}$	$O(\frac{N}{2n}+\frac{n}{2})$	$O(\frac{N}{n}+n)$
Derby-Klassen [21]	1.5	3	3	3	3
Super-compact [16]	1.25	6	$O(1)$	$O(1)$	O(1)
AQM [66, 61]	$2-\frac{1}{N}$	6	$O(1)$	$O(1)$	O(1)
$n\times n$ **Hybrid+**	$1+\frac{1}{n^2}$	$3+$ $2\mathscr{BK}_{n^2}$	$\frac{2}{n}+\frac{1}{n^2}+$ $\frac{n+1}{n}\overline{\mathscr{BK}}_{n^2}$	$O(\frac{n}{2})$*	$O(n)$
Low-ratio mapping [16, 61]	$1+\frac{1}{k}$	$O(k)$	$O(k)$	$O(k)$	$O(k)$

Table 2.1 Comparison of the performance (on an $N\times N$ square-lattice fermionic interaction graph) of select families of fermion-to-qubit mappings including state-of-the-art alongside our novel mapping $n\times n$ Hybrid and Hybrid+ mappings for $n\ll N$. $\mathscr{BK}_m$ and $\overline{\mathscr{BK}}_m$ represent the $O(\log m)$ maximum and average Pauli-weight of fermionic operators under the Bravyi-Kitaev mapping on m fermionic modes. For each hybrid mapping, n is fixed. For $n=1$ the Hybrid mapping is equivalent to the Jordan-Wigner mapping, and the Hybrid+ mapping is equivalent to AQM. For $n=N$ both are equivalent to the Bravyi-Kitaev mapping. Intermediate values of n combine the strengths of these mappings to give improved performance. The optimal choice of n is dependant upon N and must be found numerically (Figures 2.12,2.13). For $N=64$, the optimal choice is $n=8$. The 8×8 Hybrid mapping gives 98% shorter Pauli-strings than the Jordan-Wigner mapping, and the 8×8 Hybrid+ mapping only requires 1.016 qubits per fermion. This is 94% fewer ancillae than the super-compact mapping (which requires the fewest ancillae of all known local mappings). *These values are estimated using numerical evaluation.

2. Assign a *cell number* (c_i) to each of these cells in a Z pattern starting with the top-left most cell (illustrated by Figure A.1.1 in the Appendix). This enumeration reflects the Jordan-Wigner-esque properties of the mapping.

3. Within these cells assign each fermion a *mode number* (f_i) from 1 to $n \times n$ in a Z pattern starting with the top-left most fermion. We can now refer to the i-th fermion in the graph via the coordinate (c_i, f_i).

4. Repeat this procedure on a square qubit lattice of the same size so for every fermion there is an equivalent qubit labeled with the same coordinate

5. For each interaction edge in your fermion interaction graph between (c_1, f_1) and (c_2, f_2) generate the Pauli-string representation:

   (a) Generate the update U_i, flip F_i, and parity P_i sets for f_1 and f_2 independently in a Fenwick tree of size $n \times n$
   $U_i = (c_i, \text{GENERATEUPDATESET}(f_i, n \times n))$
   $F_i = (c_i, \text{GENERATEFLIPSET}(f_i))$
   $P_i = (c_i, \text{GENERATEPARITYSET}(f_i))$

   (b) Add the roots of every cell lower in the Z pattern than c_1 and c_2 to their respective parity sets.
   FOR ALL $c_j < c_1$ ADD $(c_j, n \times n)$ TO P_1
   FOR ALL $c_j < c_2$ ADD $(c_j, n \times n)$ TO P_2

   (c) Pauli-string representation is given by:

$$
\begin{aligned}
a_1 a_2^\dagger &\to X_{U_1} \sigma_1^- Z_{P_1/F_1} X_{U_2} \sigma_2^+ Z_{P_2/F_2} \\
&= \frac{1}{4} \left(\bigotimes_{j \in U_1} X_j \otimes X_{(c_1,f_1)} \otimes \bigotimes_{j \in P_1} Z_j + ß \bigotimes_{j \in U_1} X_j \otimes Y_{(c_1,f_1)} \otimes \bigotimes_{j \in P_1/F_1} Z_j \right) \\
&\left(\bigotimes_{j \in U_2} X_j \otimes X_{(c_2,f_2)} \otimes \bigotimes_{j \in P_2} Z_j - i \bigotimes_{j \in U_2} X_j \otimes Y_{(c_2,f_2)} \otimes \bigotimes_{j \in P_2/F_2} Z_j \right) \\
&= \frac{1}{4} \bigotimes_{j \in (U_1 \oplus U_2)} X_j \bigotimes_{j \in (P_1/F_1 \oplus P_2/F_2)} Z_j \otimes \left(X_{(c_1,f_1)} \otimes \bigotimes_{j \in F_1} Z_j + i Y_{(c_1,f_1)} \right) \left(X_{(c_2,f_2)} \otimes \bigotimes_{j \in F_2} Z_j - i Y_{(c_2,f_2)} \right)
\end{aligned}
$$

It is important to note that while we have presented the Hybrid mapping in terms of the even fermionic operators (keeping inline with the Hybrid+ mapping which can only represent even fermionic operators), it can also admit a representation of the whole fermionic space. Figures 2.1 to 2.5 illustrate in detail how the odd fermionic operators are mapped, and it is clear how to construct the even fermionic operators by multiplying these together.

Hybrid+ mapping algorithm

Input: *$N \times N$ Fermion Square-Lattice Interaction Graph, Cell Shape $n \times n$, $n \times n = 2^k$* for some integer k, n is a factor of N

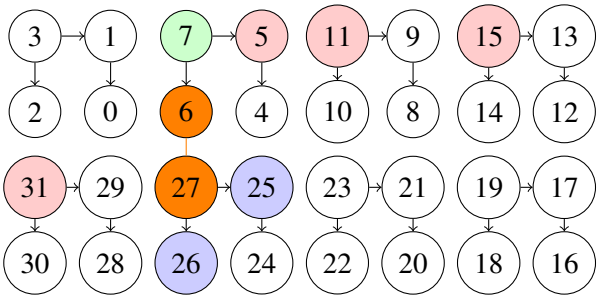


Fig. 2.1 Hybrid mapping with 2×2 cells showing the different sets of qubits involved in a fermionic operation between mode 6 and 27. Orange = targets, Blue = Flip set, Red = Parity set, Green = Update set

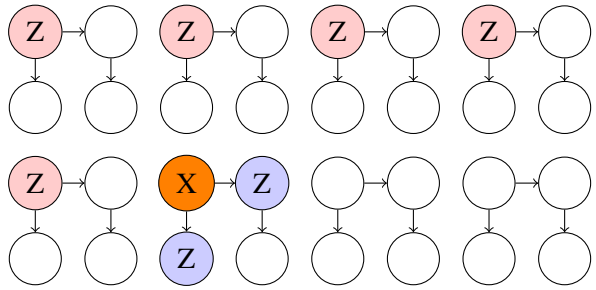


Fig. 2.2 Hybrid mapping with 2×2 cells showing the qubit operator representation of $\frac{1}{2}(a_{27} + a_{27}^{\dagger})$. Orange = target, Blue = Flip set, Red = Parity set

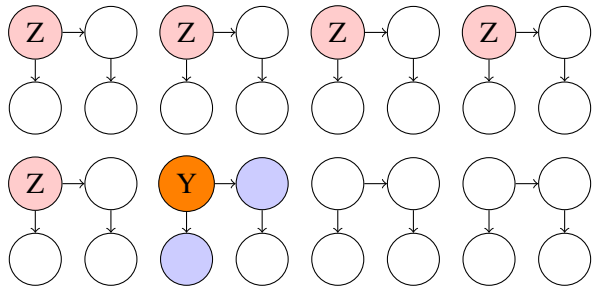


Fig. 2.3 Hybrid mapping with 2×2 cells showing the qubit operator representation of $\frac{i}{2}(a_{27} - a_{27}^{\dagger})$. Orange = target, Blue = Flip set, Red = Parity set

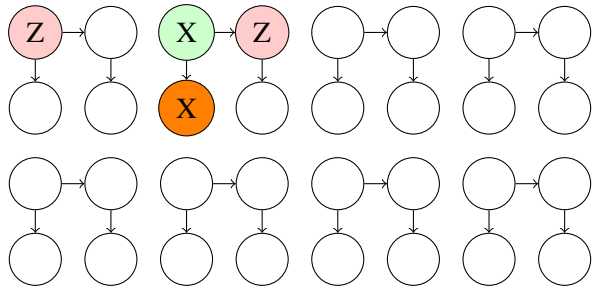


Fig. 2.4 Hybrid mapping with 2×2 cells showing the qubit operator representation of $\frac{1}{2}(a_6 + a_6^{\dagger})$. Orange = target, Red = Parity set, Green = Update set

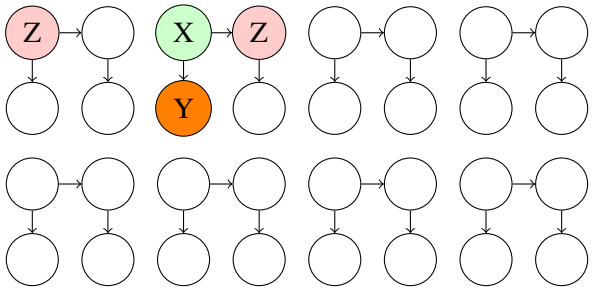


Fig. 2.5 Hybrid mapping with 2×2 cells showing the qubit operator representation of $\frac{i}{2}(a_6 - a_6^\dagger)$. Orange = target, Red = Parity set, Green = Update set

Output: *Pauli-strings corresponding to interaction edges, Enumerated square-lattice of qubits plus required ancillas, Preprocessing entanglement gates*

Summary: This algorithm essentially extends the Jordan-Wigner portion of the Hybrid mapping scheme in the same way AQM extends the Jordan-Wigner mapping. We add in an ancilla per tree and then perform entangling operators upon the roots of the trees and the ancillas such that particular stabiliser operations are constructed. We can then use these stabilisers to shorten non-local strings on the roots of the trees to a single operation on an ancilla.

Algorithm:

1. Perform steps 1-4 of the Hybrid mapping algorithm, but alternatively assign the *cell number* (c_i) in a S pattern (illustrated by Figure A.1.2 in Appendix A.1.2).

2. For every cell c_i, add one ancilla to the qubit enumeration scheme. This ancilla qubit is labelled $(c_i, n \times n + 1)$ and only needs to be connected to the root of the cell $(c_i, n \times n)$ and the ancillas of the neighbouring cells.

3. Prepare a stabiliser for each ancilla containing the string of Z gates involved in performing a fermionic operation between the roots of the current cell and the cell vertically above. (A detailed description of these stabilisers is given in Appendix A.1.2 Figure A.1.3). The stabilisers are created using the method described in [61] by applying the following series of entangling gates:

   (a) Apply a Hadamard gate to every ancilla

   (b) Entangle the edges of the rows by applying either of the gates controlled by the ancilla C shown in Figure 2.6 depending on the direction of the S-pattern on the row.

   (c) Move along each row applying either of the gates controlled by ancilla C shown in Figure 2.7 depending on the direction of the S-pattern on the row.

   This technique exploits the newly created stabiliser of the previously entangled ancilla to cancel out most of the gates required to entangle the next ancilla in the row.

4. For each interaction edge in your fermion interaction graph between (c_1, f_1) and (c_2, f_2) generate the Pauli-string representation:

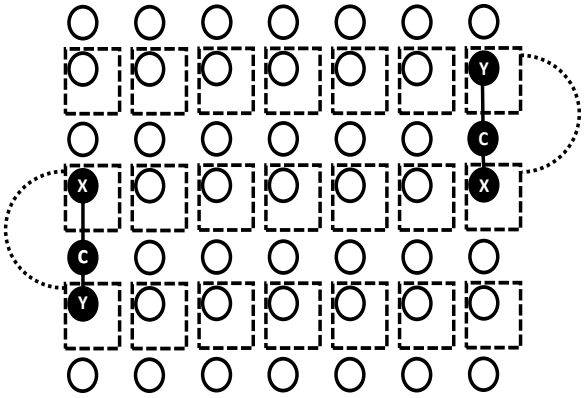


Fig. 2.6 Gates performed on edges of rows conditional upon the ancillas at the end of each row labeled with C. The dashed boxes represent the cells, whilst the circles inside them represent the roots of each cell, and the circles without boxes represent the ancillas.

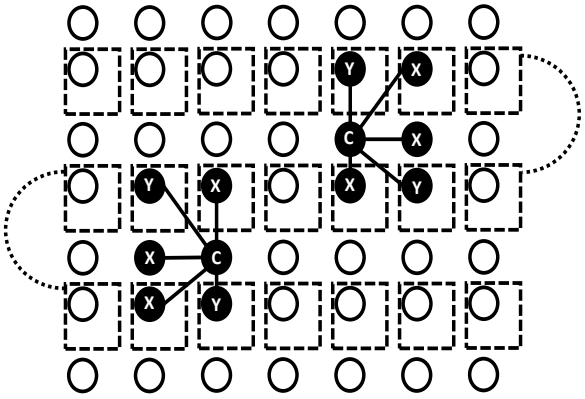


Fig. 2.7 Examples from the two (for odd/even rows) series of gates applied conditional upon the ancillas labeled with C. Each element in the series is identical to the previous but shifted one row to the left (odd rows) or right (even rows). The first element in each series is the 2nd rightmost (odd rows) and 2nd leftmost (even rows).

(a) Perform steps 5a)-c) of the Hybrid mapping algorithm

(b) We will only consider physical (even) fermionic operations, i.e. those which preserve the fermion number parity. For example:

$$a_1 a_2^\dagger + a_2 a_1^\dagger \to X_{U_1}\sigma_1^- Z_{P_1/F_1} X_{U_2}\sigma_2^+ Z_{P_2/F_2} + X_{U_2}\sigma_2^- Z_{P_2/F_2} X_{U_1}\sigma_1^+ Z_{P_1/F_1}$$
$$\to \frac{1}{2} \bigotimes_{j\in(U_1\oplus U_2)} X_j \bigotimes_{k\in(P_1/F_1\oplus P_2/F_2)} Z_k$$
$$\left(X_{(c_1,f_1)} \otimes X_{(c_2,f_2)} \bigotimes_{k\in F_1} Z_k \bigotimes_{j\in F_2} Z_j + Y_{(c_1,f_1)} \otimes Y_{(c_2,f_2)} \right)$$

In what follows we only need to consider how these operators act on the roots of the cells $(c_i, n\times n)$, as the stabiliser only deals with the cell roots. As $n\times n = 2^k$ each cell will contain a connected Fenwick tree, so each root will either be in the update set or be one of the targets. Therefore, each root is either acted on by $X_{(c_i,f_i)}$ or $X_{(c_t,f_t)} \otimes X_{(c_i,f_i)} \bigotimes_{k\in F_t} Z_k \bigotimes_{j\in F_i} Z_j + Y_{(c_t,f_t)} \otimes Y_{(c_i,f_i)}$.

(c) We need to modify every string so they preserve the stabilisers. For each stabiliser, add a Z gate (acting on the associated ancilla) to every string that anti-commutes with the stabiliser. We are also able to shorten long S pattern Pauli strings by multiplying by appropriate stabilisers. Figure 2.8 demonstrates how the Hybrid+ mapping modifies a Pauli-string representing a local fermionic operation in the Hybrid mapping scheme. Further examples are illustrated by Figures A.1.4 and A.1.5 in Appendix A.1.2.

If neither of the target fermions are the roots of their cell then the fermionic operation (on the level of cell roots) will simply be $XZ\ldots ZX$. However, if one of them is a root it will also include terms such as $XZ\ldots ZY$ and $YZ\ldots ZX$ (and potentially $YZ\ldots ZY$ if they are both roots).

2.1.2 Algorithm analysis

Theoretical Average Pauli-weight of $n\times n$ Hybrid on an $N\times N$ lattice

There are two primary contributions to the Pauli-weight of a fermionic operation in the Hybrid scheme:

1. Local (Bravyi-Kitaev type) gates acting on qubits within the target cells
2. Non-local (Jordan-Wigner type) gates acting on the roots of the cells between the two target cells in the Z-pattern

The first contribution scales with $O(\log(n^2))$, whereas the second contribution scales with $O(N)$.

We can analytically calculate the second contribution:

- Any interactions within a single cell make no contributions here

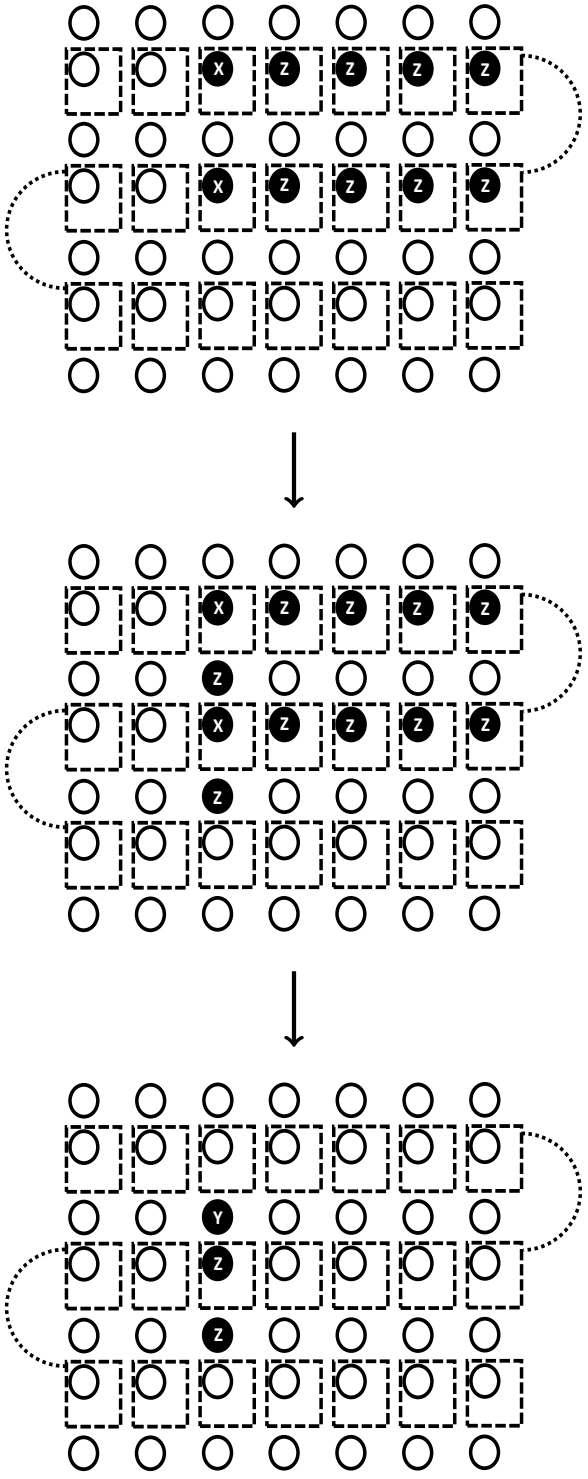


Fig. 2.8 Diagram showing how the Pauli-string representing a vertical hopping term (top) is modified by the Hybrid+ mapping algorithm. First, Z gates are added to all the ancillas with associated stabilisers that anti-commute with the string (middle). Then, a stabiliser shown in Figure A.1.3 is applied canceling out a large portion of the string to give the final local operation (bottom)

- Horizontal interaction terms between cells only contribute the root of one the target cells. However, as this root will already either be the target or in the update set we can ignore this contribution.
- Vertical interaction terms between qubits in different cells contribute $\frac{N}{n} - 1$. Therefore, the total contributions from each row is:

$$\sum_{i=1}^{N} \frac{N}{n} - 1 = N\left(\frac{N}{n} - 1\right)$$

  Therefore, as there are $\frac{N}{n}$ rows the total of the second contribution is:

$$\frac{N^2}{n}\left(\frac{N}{n} - 1\right)$$

  As there are $2N(N-1)$ interactions in the whole lattice this gives the following contribution to the average Pauli-weight:

$$\frac{N}{2(N-1)n}\left(\frac{N}{n} - 1\right)$$

Therefore, the average Pauli-weight of $n \times n$ Hybrid on an $N \times N$ lattice is:

$$\frac{N(N-n)}{2(N-1)n^2} + O(\log n)$$

and so for large lattices:

$$\frac{N}{2n^2} + O(\log n)$$

When $N \gg n$, the Hybrid mapping produces n^2 times smaller Pauli-strings than the Jordan-Wigner mapping.

Theoretical Average Pauli-weight of $n \times n$ Hybrid+ on an $N \times N$ lattice

Similarly to the Hybrid mapping, there are two primary contributions to the Pauli-weight of a fermionic operation:

1. Local (Bravyi-Kitaev type) gates acting on qubits within the target cells
2. Local (AQM type) gates acting on ancilla qubits

The first contribution scales with $O(\log(n^2))$, whereas the second contribution scales with $O(\frac{1}{n})$ (as the ratio of inter-cell/intra-cell interactions decreases as n increases). Therefore, as N (and therefore the number of fermionic modes) increases the average Pauli-weight will remain constant.

Varying impact of connectivity

The limitations of the underlying qubit architecture has a large impact on the performance of various mappings. In this section, we demonstrate the advantages our mappings give over existing schemes when

used on a quantum computer with the common square-lattice connectivity. The connectivity constraints impact mappings by requiring SWAP gates to perform entangled operations between physically non-local qubits. In our analysis, we compare the number of qubits acted upon by a Pauli or SWAP gate during a fermionic operation (the *range*). Below we consider the varying impact of connectivity upon different mappings.

Jordan-Wigner Mapping. The Jordan-Wigner mapping only requires a linear connectivity mapping, so it requires no additional gates to use on a square-lattice of qubits. However, it has been shown that an optimal choice of enumeration can give a constant improvement in the Pauli weight [18].

Bravyi-Kitaev Mapping. The Bravyi-Kitaev mapping requires a very high level of connectivity. In order to avoid any additional gates one must be able to embed an $\log N$ degree graph (the Fenwick tree) within the qubit architecture. As a square-lattice of connectivity has degree 4 it is not possible to embed a Bravyi-Kitaev mapping of more than 16 fermionic modes. Therefore, for Bravyi-Kitaev mappings on larger systems neighbouring Fenwick tree nodes will not be connected, so additional SWAP gates are required to perform entangled gates on these qubits.

We found that simulating these additional SWAP gates drastically reduced the performance of the Bravyi-Kitaev mapping. This is theoretically supported by considering the scaling of the largest range interactions. Consider the fermionic operation between the fermions corresponding to qubits which are physically located at opposite ends of a row (or column), this operation must include these qubits as they are the targets. To connect just these two qubits at least N qubits must be involved in the operation. Our unoptimized Z-pattern enumeration for the Bravyi-Kitaev mapping includes non-local interactions such as these, and hence the average range scales with $O(N)$. It is feasible that an optimal Bravyi-Kitaev enumeration could be found that scales more favourably. However, this is a very hard problem as the number of possible enumeration schemes grows factorially with lattice size and there is no known solution.

Hybrid mapping. The main strength of this mapping is its performance in the real-world scenario of a square-lattice connectivity. It combines the Jordan-Wigner mappings connectivity flexibility with the Bravyi-Kitaev mappings increased locality. The average interaction qubit count contribution from within the Bravyi-Kitaev cells scales as above with $O(n)$. While the Pauli operations on the roots of the cells along the S-pattern are joined up with SWAP gates making the contribution per row from vertical interactions:

$$\sum_{i=1}^{N}(N-1) = N(N-1)$$

Therefore, as there are $\frac{N}{n}$ rows and $2N(N-1)$ interactions in the whole lattice, the contribution to the average range is:

$$\frac{N}{2n}$$

Therefore, the average range scales with $O(n+\frac{N}{2n})$. So for $N \gg n$, this mapping has n times lower average range than both the Bravyi-Kitaev and Jordan-Wigner mappings.

Hybrid+ mapping. This algorithm requires a slightly higher level of connectivity. Our formulation has a second square-lattice of ancilla qubits connected to the original at the roots of the cells. It could also

be formulated by instead choosing one qubit in each cell of the original square lattice to be an ancilla. However, we chose our formulation as it produces a lower range, is simpler to consider, and still is a realistic qubit architecture.

The Hybrid+ mapping retains a constant average Pauli-weight when considering the limitations of connectivity. This is due to the only rising contribution coming from the Bravyi-Kitaev mapping within the target cells which should scale with roughly $O(n)$, and for a given mapping n is constant.

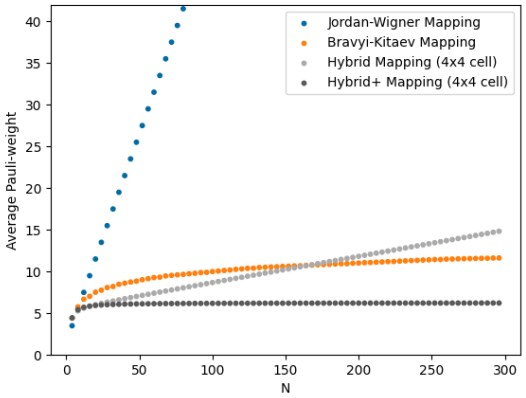


Fig. 2.9 The average Pauli-weight of fermionic operations on an $N \times N$ lattice against N when using a given mapping.

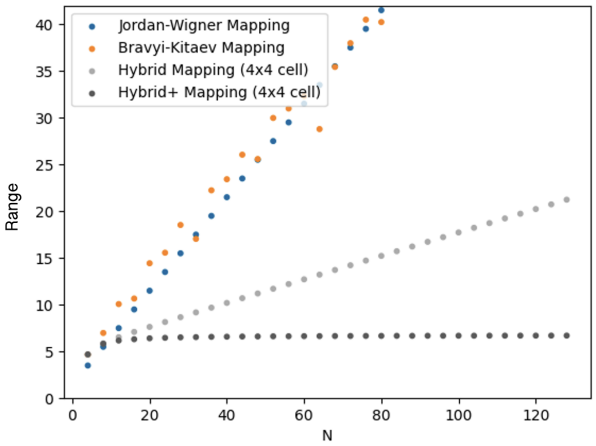


Fig. 2.10 The average range of each fermionic operation on an $N \times N$ lattice against N when using a given mapping in the context of a square-lattice qubit connectivity.

Numerical Analysis

We have considered the locality of each algorithm by computing the average range on a $N \times N$ square lattice. When investigating the case of a 4×4 cell, we found that the theoretical predictions held; the

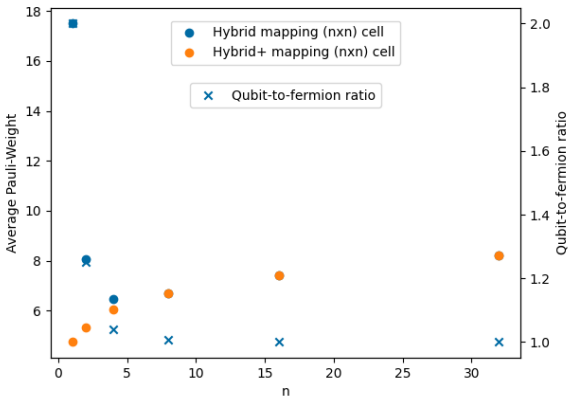


Fig. 2.11 The average Pauli-weight of a fermionic operation on an 32×32 lattice when using an $n \times n$ Hybrid/Hybrid+ mapping against n. The crosses denote the qubit-to-fermion ratio of the $n \times n$ Hybrid+ mapping for each choice of n.

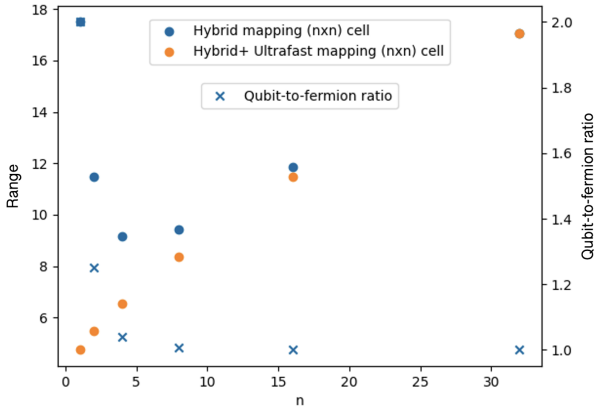


Fig. 2.12 The average range of each fermionic operation on an 32×32 lattice when using an $n \times n$ Hybrid/Hybrid+ mapping against n in the context of a square-lattice qubit connectivity.The crosses denote the qubit-to-fermion ratio of the $n \times n$ Hybrid+ mapping for each choice of n.

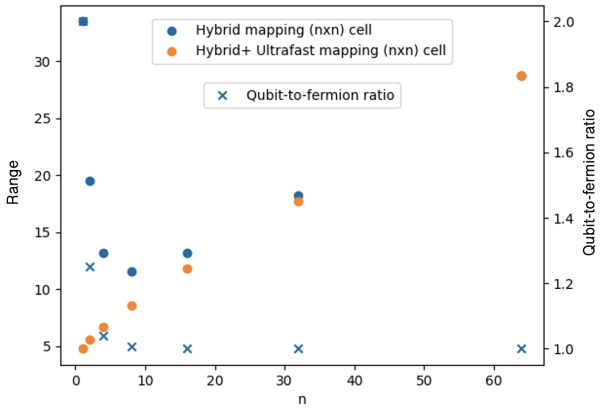


Fig. 2.13 The average range of each fermionic operation on an 64×64 lattice when using an $n \times n$ Hybrid/Hybrid+ mapping against n in the context of a square-lattice qubit connectivity for $n = \{1, 2, 4, 8, 16, 32, 64\}$. These results include qubits solely acted on by SWAP gates. The crosses denote the qubit-to-fermion ratio of the $n \times n$ Hybrid+ mapping for each choice of n.

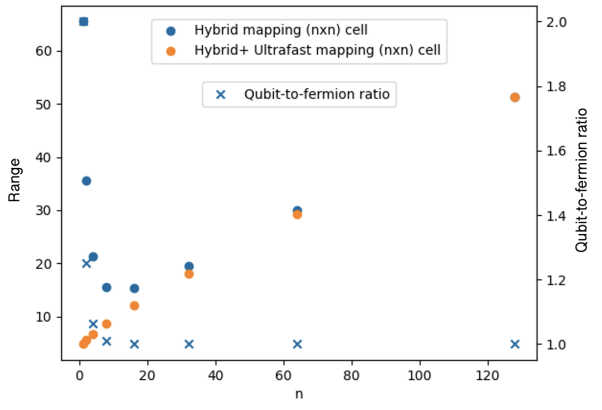


Fig. 2.14 The average range of each fermionic operation on an 128×128 lattice when using an $n \times n$ Hybrid/Hybrid+ mapping against n in the context of a square-lattice qubit connectivity for $n = \{1, 2, 4, 8, 16, 32, 64, 128\}$. These results include qubits solely acted on by SWAP gates. The crosses denote the qubit-to-fermion ratio of the $n \times n$ Hybrid+ mapping for each choice of n.

average interaction qubit count scaled with $\frac{N}{32}$ for the Hybrid algorithm and asymptotically approached a constant for the Hybrid+ algorithm (as illustrated in Figure 2.9).

We performed the same analysis in the context of an underlying square lattice connectivity. This required calculating the minimum number of SWAP gates required to connect all the qubits involved in the operation. We approximated the solution to this Steiner Tree problem using the SCIP-JACK library outlined in Appendix A.1.1. Figure 2.10 shows that the theoretical predictions hold with the average range increasing with $\frac{N}{8}$ for the Hybrid algorithm and tending to a constant for the Hybrid+ algorithm.

The largest difference evident is in the Bravyi-Kitaev mapping. When including SWAP gates, the average range involved appears to be scaling linearly rather than logarithmically. In our calculations we used a trivial Z-pattern enumeration scheme for the Bravyi-Kitaev mapping, so it is possible the performance could be improved by an optimal choice of enumeration scheme.

Therefore in the context of an underlying square lattice connectivity the 4×4 Hybrid mapping outperforms both the Jordan-Wigner and Bravyi-Kitaev mapping by a factor of 4 . Furthermore, even with all-to-all connectivity it still outperforms the Jordan-Wigner mapping by a factor of 16, and beats the Bravyi-Kitaev mapping for lattices smaller than 164×164.

We have investigated which cell sizes give optimal performance. The behaviour shown in Figure 2.11 and 2.12 matches with our theoretical predictions. For the Hybrid mapping, in the regime of $n \ll N$ the average range scales with $O(\frac{1}{2n^2})$ (for fixed N). However, this is quickly dominated by the $O(\log n)$ growth of the Bravyi-Kitaev mapping within each cell. Therefore, as the logarithm dominates for $K\frac{N}{2n^2} < \log n$ (for constant K), we hypothesis that the optimal choice of n will scale with roughly $O(\sqrt{N})$. The trend of increasing optimal choice of n with lattice size is demonstrated by the increasing optimal cell size from 4×4 to 8×8 to 16×16 when the lattice size is increased from 32×32 (Figure 2.12) to 64×64 (Figure 2.13) to 128×128 (Figure 2.14).

For the Hybrid+ mapping, there is no $O(\frac{1}{2n^2})$ regime as there are no non-local gates acting on the roots. Therefore, the behaviour is dominated by the Bravyi-Kitaev mapping within the cell. This is reflected in the $O(\log n)$ growth of average range. However, the optimal cell size is not simply the smallest (equivalent to AQM [61]). The main benefit of this mapping is in offering a direct trade-off between number of ancillas and range. As the current minimum qubit-to-fermion ratio achieved by a local mapping is 1.25 [16], we have analysed the next possible cell size of 4×4 which achieves a qubit-to-fermion ratio of 1.0625. However, using larger cell sizes requires far fewer ancilla qubits. For example, an 8×8 Hybrid+ mapping requires 94% fewer ancillas than the super-compact mapping scheme [16].

2.1.3 Discussion

We have presented two novel mappings: the Hybrid mapping achieves the shortest circuits of all known mappings without ancillas. The margin of improvement is significant. As demonstrated in our simulations of a real-world qubit architecture, it generates 75% lower range interactions than the Jordan-Wigner and Bravyi-Kitaev mappings. Additionally, as the lattice sizes increases the optimum cell size grows and the performance of the Hybrid mapping further improves.

For calculations on current or near-term quantum computers the Hybrid mapping demonstrates an improvement over the Bravyi-Kitaev and Jordan-Wigner mappings even with all-to-all connectivity. This holds for lattices up to 164×164 (Figure 2.9). Lattices larger than this would require more than 27000 qubits to simulate, which is much larger than the current largest quantum computer (433 qubits [28]). Therefore, the Hybrid mapping will likely outperform the Bravyi-Kitaev and Jordan-Wigner mappings on real devices for the foreseeable future.

The Hybrid+ mapping demonstrates a remarkable improvement on all existing local mapping schemes. It achieves similar levels of locality to the Derby-Klassen [21], Super-compact [16], and AQM [61] type mappings but expends only a small fraction of the ancillas in comparison. Our example of a 4×4 Hybrid+ mapping required 75% less ancillas than the local (super-compact) mapping with the lowest known ancilla requirements.

The Hybrid+ mapping also provides a direct and simple trade-off between locality and ancilla qubits. Increasing the cell size n increases the range linearly whilst the number of ancillas required decreases with $O(\frac{1}{n^2})$.

The approach we have taken in constructing both of these mappings need not be restricted to square-lattice qubit architectures. It should be possible to reproduce similar mappings for any architecture and corresponding fermionic interaction graph that admit a cellular decomposition.

We have considered a fermionic interaction graph which coincides with our architecture. It would be interesting to explore mapping an arbitrary fermionic interaction graph onto a given architecture using a similar Hybrid construct of connected Bravyi-Kitaev cells. In particular, we would like to understand how this mapping would translate into 3D, and whether this would require a 3D qubit connectivity or if cubic cells can be mapped to a planar grid efficiently.

In our analysis, we found connectivity constraints significantly hampered the Bravyi-Kitaev type terms in the mapping. Further work could explore this phenomenon in more detail. It is possible that an optimal choice of fermionic enumerations following the optimization techniques of [18] could significantly reduce the impact of the connectivity constraints.

A lesser-known but equally remarkable role of fermion-to-qubit mappings is their role as duality transformations. The Jordan-Wigner transform [39] can be regarded as an example of such a duality transform [34]. It maps spin Hamiltonians (such as XXZ chain) to simple fermionic systems which can be investigated by classical quantum chemistry techniques or by quantum means [50, 11]. Our two new families of fermion-to-qubit mappings could be interpreted as efficient duality transformations. Therefore, an interesting application of these hybrid mappings is the investigation of complex Hamiltonians.

2.2 Local Jordan-Wigner transformations and topological sectors on the torus

Quantum wavefunctions are described on a projective space, and this allows a system of indistinguishable particles to be described by anti-symmetric wavefunctions, a.k.a. fermions. By imposing a 1-dimensional ordering to a collection of fermions, Jordan and Wigner [39] demonstrated how to endow the fermions with a tensor product structure. This was done by introducing a duality mapping between the fermions and a system of spin $1/2$ particles (i.e. qubits). However, this original mapping only preserves locality for 1-D systems. The concept of flux attachment [36, 46, 30, 35, 26] suggested the existence of higher-dimensional analogues of this Jordan-Wigner transformation preserving locality in arbitrary dimensions and in the thermodynamic limit; the first explicit constructions of such a mapping in 2 spatial dimensions was obtained in [7, 66, 4]. A system of fermions located on the vertices of a square lattice was mapped to a system of qubits living on the edges of that graph. The fact that this construction yields twice as many qubits as fermionic modes was compensated by the fact that the new system exhibits gauge conditions on every plaquette. This work was followed by many other papers [57, 65, 31, 67, 33, 59, 62, 38, 3, 37, 22, 19, 49, 17, 51] that improved the qubit-to-fermion ratio and/or the maximum support in the spin representation of a single fermionic hopping term. The modern point of view is that the fermion to qubit mapping is a duality transformation obtained by gauging the global Z_2 parity symmetry of the fermions [15]; in such a mapping, the fermion matter can then be disentangled by a local fermionic quantum circuit [69]. However, many questions remained open: it was not clear how to modify the mappings when dealing with odd charge sectors especially when dealing with periodic and twisted boundary conditions on a torus. In this paper, we show how those problems can be solved by making use of the formalism of graded tensor networks [10, 20] and more specifically of projected entangled pair operators which explicitly encode the gauge transformations [29, 60].

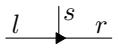
This section is organized as follows. We start by defining the graded projected entangled pair operator (PEPO) for the fermion to qubit mapping on arbitrary graphs. Subsequently, we revisit the 1-D Jordan-Wigner transformation for a (anti-)periodic ring of fermions in the even and odd charge sectors in terms of matrix product operators (MPOs). Next, we show how this mapping can readily be extended to the 2-dimensional torus, and pay special attention to the intriguing interplay between charge sectors and boundary conditions. Controlling the possible boundary conditions turns out to be important for state of the art simulations of e.g. the Hubbard model [68]. Hence our results can be utilised to improve fermionic quantum simulation by reducing the finite size effect through the introduction of (anti-)periodic boundary conditions.

The larger motivation behind our work is to reduce the entanglement of fermionic ground states by mapping them to a spin setting, following recent work [44]. Our PEPO introduces an extensive number of symmetries on the spin side, resulting in a highly degenerate spin ground state corresponding to a unique fermionic ground state. Within this degenerate space, a state might exist with significantly lower entanglement that breaks all the introduced symmetries. This lower-entanglement state is more amenable

to tensor network methods, enabling efficient simulation of highly entangled fermionic systems through our construction.

Construction—Given an arbitrary graph $G = (V,E)$, we construct our PEPO by placing fermionic Z_2 graded $(d+1)$-degree vertex tensors [10] at every d-degree vertex of G and a graded GHZ tensor on every edge of G.

The GHZ tensor is defined as:



$$:= \sum_c \langle c|_s (c|_l (c|_r .$$

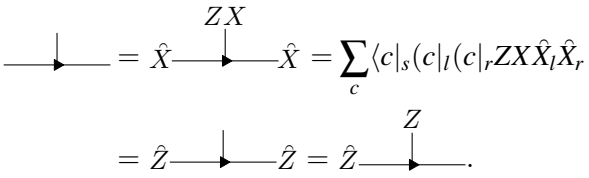
where l and r label the left and right virtual fermionic legs of the tensor and s labels the physical spin leg. Every tensor must be drawn with an arrow which dictates what is left and right, but as we work on arbitrary graphs, no spin structure is needed. Curved brackets will be used to denote fermionic degrees of freedom, while the usual bra-ket notation will be used for the qubit degrees of freedom. We can consider curved brackets to be defined in terms of creation operators acting on the vacuum with $|1) = c^\dagger|\Omega), |0) = |\Omega)$ and hence:

$$|a)|b) = (-1)^{ab}|b)|a), \quad \mathrm{Tr}(|a)(b|) = (-1)^{ab}.$$

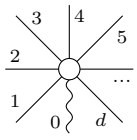
Throughout this work we refer to the fermionic $\hat{X}$ and $\hat{Z}$ operators which we define as:

$$\hat{X} = \sum_a |a+1)(a|, \quad \hat{Z} = \sum_a (-1)^a |a)(a|.$$

We study the symmetries given by applying products of these operators to the GHZ tensors:



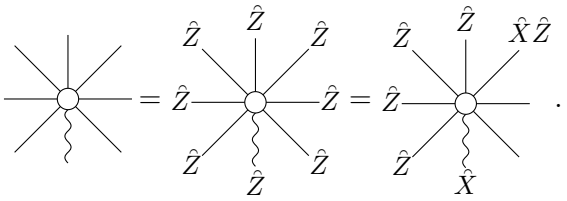
When writing symmetries in this graphical notation, we require that the operators applied are ordered with the same internal ordering of the legs which they apply to. The vertex tensor is defined as:



$$:= \sum_{\substack{\{a_i\} \\ \sum_i a_i = 0 \,(\mathrm{mod}\, 2)}} |a_d) \ldots |a_1)|a_0),$$

where a_0 labels the physical fermionic leg of the tensor and $a_1, \ldots, a_d$ label the virtual fermionic legs which will be contracted with the GHZ tensor; the ordering is important and must be specified on every vertex. In this work we always using the ordering of clockwise starting at the physical fermionic leg. Any choice of orderings will be valid, however, changing the internal ordering will lead to different mappings.

With this in mind we consider the symmetries of the vertex tensors:



The constructed PEPO will map fermionic operators acting on the vertices of G to spin operators acting on the edges of G. We represent this (before contraction of the virtual legs) as $\langle s|\,(E||V_v)|V_p)$ where $\langle s|$ are the physical spins, $(E|$ are the virtual fermions on the GHZ tensors, $|V_v)$ are the virtual fermions on the vertex tensors and $|V_p)$ are the physical fermions. This mapping will be local in the sense that even fermionic hopping terms acting along edges of the graph G will map to spin operators with a maximum Pauli weight $2\Delta-1$ (where Δ is the maximum degree of G).

As all tensors involved in the PEPO have an even parity, a tensor network diagram uniquely specifies the mapping, as long as arrows are put on all edges and an internal ordering of vertex tensors is chosen. Note that defining a fermionic operator (e.g. $\hat{O}=a_j^\dagger a_i$) requires a linear ordering for the Hilbert space it acts upon (e.g. $j<i$). A change to this ordering (e.g. to $i<j$) incurs a global minus sign on the fermionic operator giving a new definition of the same operator ($\hat{O}=-a_j^\dagger a_i$) acting on a different Hilbert space. As our mapping is not affected by the choice of linear ordering it incurs no sign change upon reordering (it will always map $a_j^\dagger a_i$ to O). Therefore, whether $\hat{O}$ is mapped to $+O$ or $-O$ depends on the linear ordering chosen for the fermionic Hilbert space being mapped.

Using the symmetry properties of our tensors before contraction, fermionic operators acting on the physical fermionic degrees of freedom (O_{pf}) are transformed to to fermionic operators acting on the virtual fermionic legs of our vertex tensors ($O_{vf(ver)}$). This is equivalent to acting with the same operators on the virtual fermionic legs of our edge tensors ($O_{vf(edg)}$), which then transform them to spin operators acting on the physical spin degrees of freedom (O_s):

$$\begin{aligned}
O_{pf}\,\langle s|\,(E||V_v)|V_p) &= \langle s|\,(E|O_{pf}|V_v)|V_p)\\
&= \langle s|\,(E|O_{vf(ver)}|V_v)|V_p)\\
&= \langle s|\,(E|O_{vf(edg)}|V_v)|V_p)\\
&= \langle s|\,O_s(E||V_v)|V_p)\\
&= \langle s|\,(E||V_v)|V_p)O_s.
\end{aligned}$$

The symmetries of our tensors reveal that the theory at the side of the spins is a gauge theory, with a gauge condition for every possible cycle in the graph. The number of independent gauge conditions is given by the circuit rank r of G (the number of independent cycles in G). As $r=|E|-|V|+1$, this mapping is an isomorphism from the even fermionic sector to the $+1$ joint eigenspace of the gauge conditions on the spin side. On a torus two of these independent gauge conditions correspond to non-contractible loops;

these gauge conditions can be lifted and interpreted as charge sectors, and those different charge sectors will correspond to (twisted) boundary conditions on the fermion side.

To admit a mapping of odd fermionic states we add a single odd tensor defect (fermionic $\hat{X}$ operator) to a virtual leg of the PEPO. This fermionic charge defect can be placed anywhere, and its presence can be understood as a boundary condition for the dual spin theory: a manifestation of the effect that dualities switch boundary conditions with charge sectors.

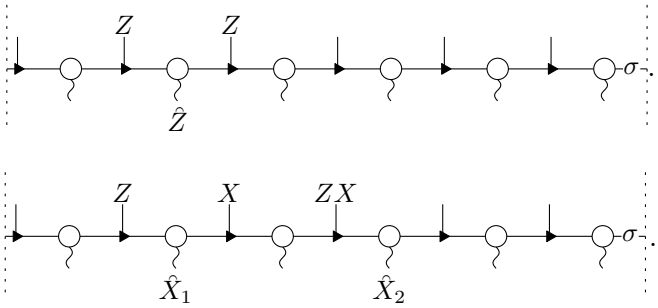
A single fermionic creation/annihilation operator will be mapped to a string of spins terminating at the odd defect. If strings of two different operators converge at any point before reaching the defect they will anticommute. If two strings never converge but instead approach the defect from opposite directions, then they will necessarily anticommute on the closest spin to the defect.

Hence, we produce the correct anti-commutation relations for the fermionic algebra.

It can be seen from the symmetries of our tensors that our construction on a general graph without a defect is an explicit PEPO representation of the BKSF mapping [7]. Therefore, due to the equivalence of 2D fermionic encodings shown by [17] it is possible to reconstruct any existing 2D fermion-to-qubit mapping through a particular choice of G, internal orderings, single qubit gates and majorana repairing. Further the inclusion of the $\hat{X}$ defect allows for the representation of tree-based mappings such as the Bravyi-Kitaev Fast Mapping [7]. We will expand on this equivalence in future work.

1D Construction—On a 1D cycle graph, our construction turns out to be equivalent to the Jordan-Wigner mapping composed with the Kramers–Wannier duality. We will use this as an example to illustrate our formalism. Further insight into how 1D boundary conditions map charge sectors can be found in Appendix D of [60].

The 1D PEPO is an MPO formed of interleaved GHZ tensors and 3-degree vertex tensors. The mappings of fermionic $\hat{Z}$ and $\hat{X}\hat{X}$ operations can be inferred from the invariance of the MPO under the following operations:



In this particular example (with σ boundary condition), a fermionic $\hat{Z}_i$ operator is mapped to a $Z_{i-1/2}Z_{i+1/2}$ term on the spin side, and a fermionic $\hat{X}_{i-1}I_i\hat{X}_{i+1}$ to $Z_{i-3/2}X_{i-1/2}(XZ)_{i+1/2}$. Different choices of boundary condition will map different topological sectors. These sectors are characterised by the charge sector under the global symmetries of each side, and the twists required for translational symmetries. Charge sectors are the eigenspaces of a symmetry operator. Twists are the modifications required to either side of a PEPO upon translation in order to recover the same PEPO.

The charge sector mapped by a boundary condition is determined by the sign produced when the MPO absorbs the given symmetry. As in the example below, it is important to account for the signs due to moving any unpaired fermionic operators at the boundaries next to each other and into the correct order to be absorbed by the vertex symmetries without additional signs.

$$\begin{aligned}
&\langle s|\,(E|\sigma_n|V_v)|V_p\rangle \prod_i X_i = \langle s| \prod_i X_i (E|\sigma_n|V_v)|V_p\rangle \\
&= \langle s|\,(E|Y_1X_2Y_3\ldots X_{n-2}Y_{n-1}X_n\sigma_n|V_v)|V_p\rangle \\
&= \langle s|\,(E|Y_1X_n\sigma_n|V_v)|V_p\rangle \\
&= \begin{cases} -\langle s|\,(E|X_n\sigma_nY_1|V_v)|V_p\rangle & \sigma \text{ is even} \\ \langle s|\,(E|X_n\sigma_nY_1|V_v)|V_p\rangle & \sigma \text{ is odd} \end{cases} \\
&= \begin{cases} -\langle s|\,(E|\sigma_n|V_v)|V_p\rangle & \sigma = I \\ \langle s|\,(E|\sigma_n|V_v)|V_p\rangle & \sigma = Z \\ \langle s|\,(E|\sigma_n|V_v)|V_p\rangle & \sigma = X \\ -\langle s|\,(E|\sigma_n|V_v)|V_p\rangle & \sigma = Y \end{cases}
\end{aligned}$$

The twist produced by a boundary condition under a translation is calculated by shifting over each physical leg above and below the MPO, and considering the necessary operators produced on the physical legs by moving the boundary condition to the new boundary. For example, translating the $\hat{Z}$ boundary condition produces a $\hat{Z}$ twist on a fermionic physical leg:

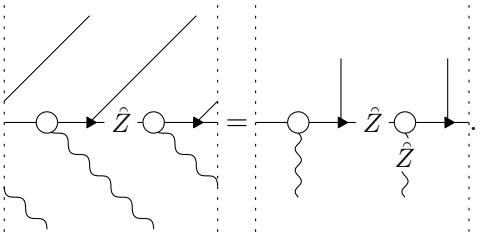


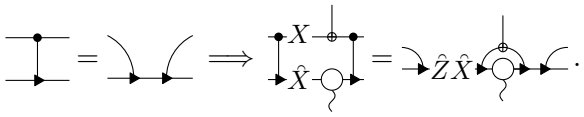
Table 2.2 illustrates how the topological sectors of the 1D cyclical graph are determined and mapped by different boundary conditions. By treating these boundary conditions as additional quantum degrees of freedom, we are able to recover a completely unitary transformation from fermions to spins. If we did not consider boundary conditions we would only have an isometry between the periodic even fermionic space and the periodic -1 spin eigenspace of $\prod_i X_i$, rather than mapping the entire fermionic Hilbert space to the entire spin Hilbert space.

We may compose our MPO with the Kramers-Wannier MPO in order to recover the canonical Jordan-Wigner transformation. The composition will necessarily evaluate to zero if the charge sectors are not matched. However, mismatched twists will give a non-zero MPO, therefore, it is necessary to chose the unique KW boundary condition which matches the twist and charge sector as seen in Table 2.2. We can

Fermions			Spins			Dual Spins		Unified
$\prod_i \hat{Z}_i$	Twist	BC 1	$\prod_i X_i$	Twist	BC 2	$\prod_i Z_i$	Twist	BC
$+1$	I	I	-1	I	Z	$+1$	Z	$\hat{Z}$
$+1$	$\hat{Z}$	$\hat{Z}$	$+1$	I	I	$+1$	I	$\hat{Z}$
-1	I	$\hat{X}$	$+1$	X	X	-1	I	$\hat{Z}\hat{X}$
-1	$\hat{Z}$	$\hat{Z}\hat{X}$	-1	X	ZX	-1	Z	$\hat{Z}\hat{X}$

Table 2.2 Mapping of topological sectors of fermions to spins on the chain dependent on the choice of boundary condition (BC 1) for the duality MPO. These sectors can then be mapped again to the dual spins by the application of a Kramers-Wannier MPO with a suitable choice of boundary condition (BC 2). The composed mapping from fermions to dual spins is equivalent to the canonical Jordan-Wigner mapping with one single boundary condition (Unified BC).

then combine these two boundary conditions as follows:

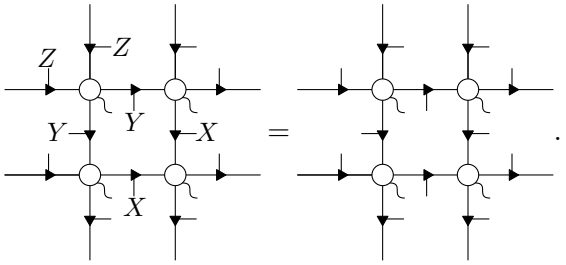


Hence, the $\hat{Z}$ and $\hat{Z}\hat{X}$ boundary conditions map both the periodic and anti-periodic boundary conditions for the even and odd sectors respectively. The I and $\hat{X}$ boundary conditions evaluate to 0 (as they correspond to mismatched charge sectors between the combined MPOs), so we can simply add them to their non-zero counterparts ($\hat{Z}$ and $\hat{Z}\hat{X}$) to obtain rank 1 boundary conditions $|0)(0|$ and $|1)(0|$. By summing these two MPOs we produce a unified MPO with boundary $(|0)+|1))(0|$ giving a unitary mapping from fermions to spins. This clarifies the mystery of why one and the same JW-mapping can deal with all four topological sectors - which is very unusual and unexpected for a duality transformation.

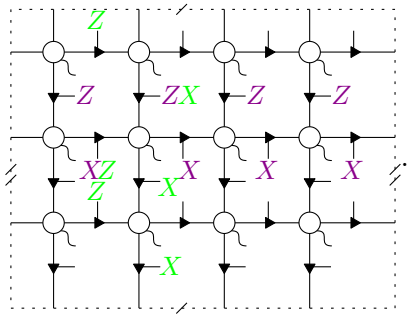
We note that using our formalism the boundary conditions are sensitive to their placement in the MPO. For example, in 1D the $\hat{Y}$ boundary condition to the right of a 3-degree vertex tensor is similar to a $\hat{X}$ boundary condition to the left of a 3-degree vertex tensor. For consistency we always refer to the boundary condition as acting to the left of the 3-degree vertex tensor.

2D Construction—For any planar graph G on an orientable manifold, there will be a natural choice for the orientations of the vertex tensors. In this section we exemplify our mapping using the square lattice. In the bulk, our PEPO is similar to the one constructed in [60] and consists out of two simple tensors, a GHZ tensor and a 5-degree vertex tensor, arranged on the edges and vertices of a square lattice. On every

plaquette, the symmetries impose a gauge condition upon the spin system:



The number of vertices is exactly equal to the number of edges minus the number of plaquettes, this is the complete set of gauge conditions, if it were not for the fact that two of them are not independent. Of course, the two missing gauge conditions are the ones winding around the torus. These are given by loops of spin operators along the horizontal (X_H, magenta) and vertical (X_V, green) directions:



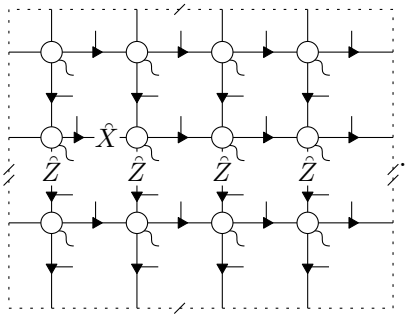
From the point of view of the fermion side, changing the charge sector on the spin side is equivalent to adding twisted boundary conditions. These boundary conditions are strings of fermionic $\hat{Z}$ operators lying along the virtual legs of the PEPO in the horizontal ($\hat{Z}_H$) and vertical ($\hat{Z}_V$) directions respectively. Table 2.3 illustrates how these boundary conditions are mapped on a torus. It is interesting to see how the four boundary conditions and two charge sectors are mapped to two boundary conditions and four charge sectors.

Boundary conditions are sensitive to their position in the tensor network, so for simplicity we will only place the X defect on horizontal virtual legs to the right of the GHZ tensor. We are free to choose which non-contractible loops are X_H and X_V. We have chosen them such that they intersect with the $\hat{X}$ defect if present, so the table matches the results for the 1D case. However, it would be equally valid to choose them to avoid the $\hat{X}$ defect which would make the spin charge sectors symmetrical for the even and odd fermionic sectors.

Fermions				Spins	
$\prod_i \hat{Z}_i$	H. Twist	V. Twist	BC	$X_H X_V$	H./V. Twist
$+1$	I	I	I	$-1-1$	I
$+1$	I	$\hat{Z}_{p_H}$	$\hat{Z}_H$	$-1+1$	I
$+1$	$\hat{Z}_{p_V}$	I	$\hat{Z}_V$	$+1-1$	I
$+1$	$\hat{Z}_{p_V}$	$\hat{Z}_{p_H}$	$\hat{Z}_V\hat{Z}_H$	$+1+1$	I
-1	I	I	$\hat{X}$	$+1+1$	$Z_v X_h / Z_h X_v$
-1	I	$\hat{Z}_{p_H}$	$\hat{Z}_H\hat{X}$	$+1-1$	$Z_v X_h / Z_h X_v$
-1	$\hat{Z}_{p_V}$	I	$\hat{Z}_V\hat{X}$	$-1+1$	$Z_v X_h / Z_h X_v$
-1	$\hat{Z}_{p_V}$	$\hat{Z}_{p_H}$	$\hat{Z}_H\hat{Z}_V\hat{X}$	$-1-1$	$Z_v X_h / Z_h X_v$

Table 2.3 Table illustrating how different boundary conditions of the PEPO on the torus map different charge sectors and produce different twists. The $\hat{Z}_{p_V}$ and $\hat{Z}_{p_H}$ twists are vertical and horizontal loops of $\hat{Z}$ operators on the physical fermions. The $Z_v X_h / Z_h X_v$ twist (with subscript v/h indicating whether the operator is applied on a vertical or horizontal leg) constitutes a single unit cell of X_H or X_V depending on whether the twist was taken in the horizontal or vertical direction respectively.

Below is an example of a PEPO with a $\hat{Z}_H$ boundary condition and an $\hat{X}$ defect, which will map between the odd fermionic algebra and the $(-1,+1)$ eigenspace of (X_H, X_V):



We have made the choice of a square lattice to illustrate the mapping of topological sectors. This choice is not necessary to reproduce the results of Table 2.3, and suitable definitions of boundary conditions/charge sectors/twists on other 2D lattices (e.g. Kagome or honeycomb) will produce identical results.

Conclusion—We have shown how boundary conditions on a torus relate to various topological charge sectors. By treating those boundary degrees of freedom as extra quantum degrees of freedom, this allows us to construct a unitary operator (as opposed to an isometry) implementing the duality transform, demonstrating that the higher dimensional JW transformation completely preserves the spectrum of this bigger theory. This is completely in line with the results in [45, 42, 43] for all possible dualities defined in terms of categorical symmetries.

Further, we have demonstrated a tensor network approach to constructing arbitrary fermion to qubit encodings. The resulting spin system is a gauge theory with a very particular type of gauge constraints. We conjecture that such gauge theories in two dimensions can only be "ungauged" with graded tensor network intertwiners such as the PEPO we have described. This highlights the difference between the 1- and the 2-D case, as fermions seem to be indispenable in the latter whereas in the 1D case we can easily bosonise the virtual legs of the MPO.

We expect our methodology could be extended to higher dimensions in line with other systematic generalisations [14, 13]. Higher dimensions would lead to even more possible boundary conditions. For example, in 3D we would expect our mapping between charge sectors and boundary conditions to be between sheets of Paulis and lines of Paulis.

End Matter on Unitarity—In this section we discuss how our mappings are made unitary by treating the defect boundary condition as an additional degree of freedom.

The 1D MPO is unitary as the Fock basis is mapped one-to-one onto a spin basis. We can represent the fermionic state $|f_0 f_1 \dots f_n)$ in the stabiliser formalism:

$$\{I, (-1)^{f_0}\hat{Z}_0, \dots, (-1)^{f_{n-1}}\hat{Z}_{n-1}, (-1)^{f_n}\hat{Z}_n\}.$$

The MPO with d odd defects maps this to:

$$\{I, (-1)^{d+1}\textstyle\prod_i X_i, (-1)^{f_0}Z_0Z_1, \dots, (-1)^{f_{n-1}}Z_{n-1}Z_n, (-1)^{f_n+d}Z_nZ_0\}, \tag{2.1}$$

where the additional $(-1)^{d+1}\prod_i X_i$ stabiliser comes from the degenerate mapping of the identity due to the symmetries of the MPO. Multiplying all the Z_iZ_{i+1} stabilisers together gives $(-1)^{\sum_i^n f_n+d}I$. As $-I$ only stabilises the empty set, there is a unique choice of boundary condition which gives a non-empty result, simplifying (2.1) to:

$$\{I, (-1)^{\sum_i^n f_i+1}\textstyle\prod_i X_i, (-1)^{f_0}Z_0Z_1, \dots, (-1)^{f_{n-1}}Z_{n-1}Z_n\}.$$

These n independent generators uniquely specify an n spin state parameterised by $(\sum_i^n f_i + 1, f_0, \dots, f_{n-1})$. Therefore, each element in n-mode Fock space is mapped to a unique n spin state, and vise versa.

We can easily extend these ideas to 2D. For a square lattice on a torus each $\hat{Z}_i$ generator is mapped to $ZZZZ_i$ (a Z on each edge adjacent to the i-th vertex), and the identity is mapped to itself, local gauge conditions around each plaquette G_i, and 2 non-local generators around the non-contractible loops (X_V, X_H):

$$\{I, (-1)^{d+1}X_V, (-1)^{d+1}X_H, G_0, \dots G_{p-1}, (-1)^{f_0}ZZZZ_0, \dots, (-1)^{f_n+d}ZZZZ_n\}$$

where there are p plaquettes and n fermionic modes (vertices). The $(-1)^{\sum_i^{n-1} f_i}ZZZZ_n$ generator can be constructed from a product of the first $n-1$ vertices so it is necessary to take $d = \sum_i^n f_i$ to stabilise a non-empty set. As there are the same number of plaquettes as vertices, there are $2n$ independent generators, and hence a unique $2n$ spin state is specified. This unique state is again parameterised by $(\sum_i^n f_i + 1, f_0, \dots, f_{n-1})$, and hence each element in n-mode Fock space is mapped to a unique $2n$ spin state.

Therefore, this is a unitary mapping between the n-mode Fock space and a gauge constrained subspace of $2n$ spins.

In the above discussion it is crucial that we were able to chose our boundary condition d dependant on the state we were mapping, or else half of the Fock space would have been mapped to nothing. Therefore, it is essential to include this boundary condition as a degree of freedom on the spin side of the mapping in order to achieve unitarity.

Chapter 3

My third chapter

3.1 First section of the third chapter

And now I begin my third chapter here . . .
And now to cite some more people **? ?**]

3.1.1 First subsection in the first section

. . . and some more

3.1.2 Second subsection in the first section

. . . and some more . . .

First subsub section in the second subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

3.1.3 Third subsection in the first section

. . . and some more . . .

First subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it and some more and some more and some more and some more and some more and some more and some more . . .

Second subsub section in the third subsection

. . . and some more in the first subsub section otherwise it all looks the same doesn't it? well we can add some text to it . . .

3.2 Second section of the third chapter

and here I write more . . .

3.3 The layout of formal tables

This section has been modified from "Publication quality tables in LaTeX*" by Simon Fear.

The layout of a table has been established over centuries of experience and should only be altered in extraordinary circumstances.

When formatting a table, remember two simple guidelines at all times:

1. Never, ever use vertical rules (lines).
2. Never use double rules.

These guidelines may seem extreme but I have never found a good argument in favour of breaking them. For example, if you feel that the information in the left half of a table is so different from that on the right that it needs to be separated by a vertical line, then you should use two tables instead. Not everyone follows the second guideline:

There are three further guidelines worth mentioning here as they are generally not known outside the circle of professional typesetters and subeditors:

3. Put the units in the column heading (not in the body of the table).
4. Always precede a decimal point by a digit; thus 0.1 *not* just .1.
5. Do not use 'ditto' signs or any other such convention to repeat a previous value. In many circumstances a blank will serve just as well. If it won't, then repeat the value.

A frequently seen mistake is to use '\begin{center}' . . . '\end{center}' inside a figure or table environment. This center environment can cause additional vertical space. If you want to avoid that just use '\centering'

Table 3.1 A badly formatted table

	Species I		Species II	
Dental measurement	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Table 3.2 A nice looking table

Dental measurement	Species I		Species II	
	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

Table 3.3 Even better looking table using booktabs

Dental measurement	Species I		Species II	
	mean	SD	mean	SD
I1MD	6.23	0.91	5.2	0.7
I1LL	7.48	0.56	8.7	0.71
I2MD	3.99	0.63	4.22	0.54
I2LL	6.81	0.02	6.66	0.01
CMD	13.47	0.09	10.55	0.05
CBL	11.88	0.05	13.11	0.04

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Appendix A

Fermion-to-Qubit Mappings

A.1 Hybrid Mappings

A.1.1 Steiner Tree Solver

In our analysis we evaluate the performance of different mappings on a square-lattice qubit architecture. Some mappings require 2-qubit gates to be performed on non-neighbouring qubits and so we need additional SWAP gates. Calculating the optimum choice of SWAP gates requires finding the minimum path length required to connect all the qubits involved in a given fermionic operation. This is an example of an NP-hard problem known as the Steiner Tree problem:

STEINER TREE PROBLEM

GIVEN AN UNDIRECTED CONNECTED GRAPH $G = (V, E)$ AND A SET $T \in V$ OF TERMINALS, FIND A TREE $S \in G$ OF MINIMUM LENGTH THAT SPANS T

In our case the set of terminals corresponds to the set of qubits involved in the given fermionic operation which we want to find the minimum path of SWAP gates between. In our calculations we use a top-ranking Steiner Tree solver SCIP-JACK [54] to find this minimum path. SCIP-JACK consists of three component parts: preprocessing, primal heuristics, and a branch-and-cut procedure.

The preprocessing uses reduction techniques to reduce the size of the graph whilst still allowing at least one solution to the reduced problem to be converted into an optimal solution to the original graph. Examples of these techniques include merging together connected vertices that share no other neighbours or discarding 1-degree non-terminal vertices.

The primal heuristic finds good or optimal solutions to the problem. In SCIP-JACK the one-phase repetitive shortest path heuristic [2] is used. This starts from a vertex and uses Dijkstra's algorithm [23] to find the shortest path to the nearest terminal. This procedure is then reiterated with the modification to Dijkstra's algorithm that the tentative distance assigned to each vertex is carried over. Once this has built up a subgraph containing all the terminals, a minimum spanning tree is constructed covering the subgraph, and any 1-degree non-terminal vertices are pruned.

The branch-and-cut procedure is the backbone of the program used to compute a lower bound and prove the optimality of the solution. First, the problem is expressed in terms of a integer programming problem by considering x_e to be binary variables representing whether edge $e \in E$ is included in the Steiner tree:

STEINER TREE PROBLEM UNDIRECTED CUT FORMULATION

Define $\delta(U) = \{\{u,v\} \in E : u \in U, v \in V/U\}$

Find min $\sum_e x_e$ under the following constraints:

1: $x(\delta(W)) \geq 1$ for all $W \in V$, $0 < |W \cap T| < |T|$

2: $x_e \in \{0,1\} \forall e \in E$

Linear programming is used to initially optimise the solution when considering only a subset of these constraints providing a lower bound on an optimal solution. The primary heuristic is then run with edge costs given by $1 - x_e$ to find a feasible solution (reasonably similar to the LP solution) providing an upper bound on the optimal solution. Then, the algorithm considers the restraints ("cuts") that the current solution violates and adds them to the subset of constraints in the linear programming model. This model is repeatedly solved and expanded until the subset of constraints has grown sufficiently. At this point, if the solution is integral then we have found an optimal solution, otherwise the algorithm "branches" by creating two new sub-problems (nodes) each with an additional constraint on one of the variables (x_e) that forces it to be either 0 or 1 in each problem respectively. The process is then repeated again on each of these nodes (potentially with more constraints added). If for any node the optimal solution exceeds any of the heuristic solutions found, the node is discarded (pruned).

A.1.2 Visual guide to Hybrid mappings

In this section we provide a series of diagrams to further elaborate and explain our hybrid mapping algorithms.

Figures A.1.1 and A.1.2 respectively describe the Jordan-Wigner mapping over the roots of Bravyi-Kitaev cells in the Hybrid and Hybrid+ mapping schemes.

Figure A.1.3 illustrates the family of stabilisers utilised by the Hybrid+ mapping scheme. This is identical to the stabilisers found in [61].

Figures A.1.4 and A.1.5 demonstrate how the Hybrid+ mapping scheme shortens different Pauli-strings.

In the figures below: the dashed boxes represent the cells, whilst the circles inside them represent the roots of each cell, and the circles without boxes represent the ancillas.

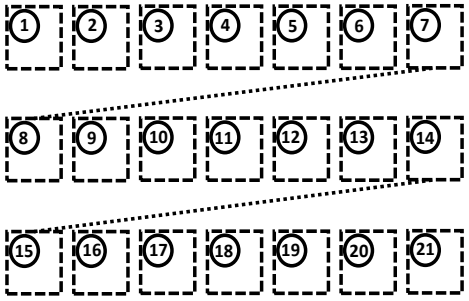


Fig. A.1.1 Diagram illustrating how the *cell number* is assigned in a Z pattern to each cell in the Hybrid mapping.

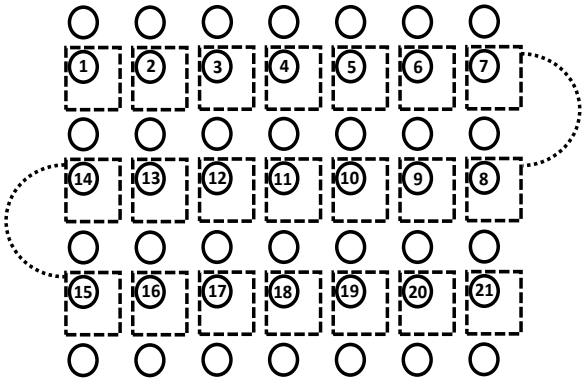


Fig. A.1.2 Diagram illustrating how the *cell number* is assigned in a S pattern to each cell in the Hybrid+ mapping. The unnumbered circles represent the ancillas each connected to the cell directly below them.

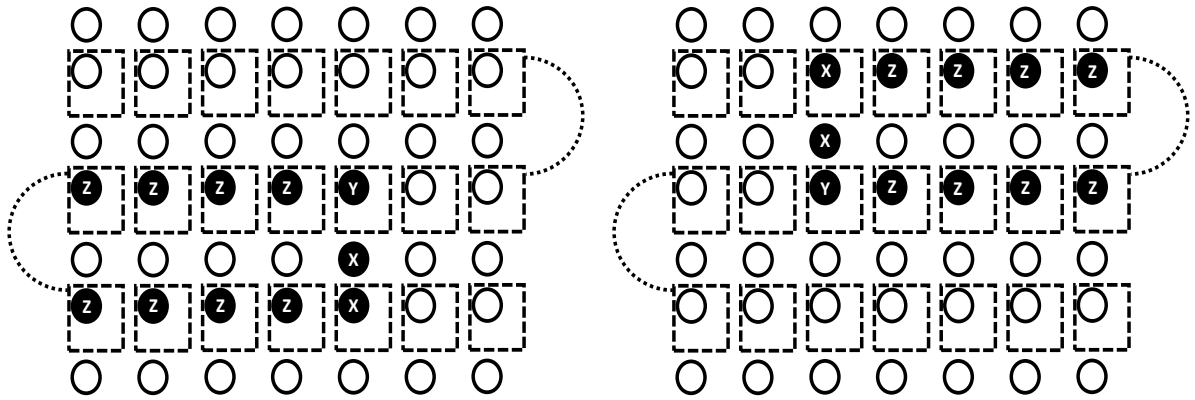


Fig. A.1.3 Examples from the family of stabilisers created by applying the series of gates illustrated in 2.6 and 2.7. The family consists of a stabiliser for every ancilla which is a string of Z gates that follow the line of the S-pattern with either XXY or YXX gates surrounding the ancilla depending on whether the row is odd or even.

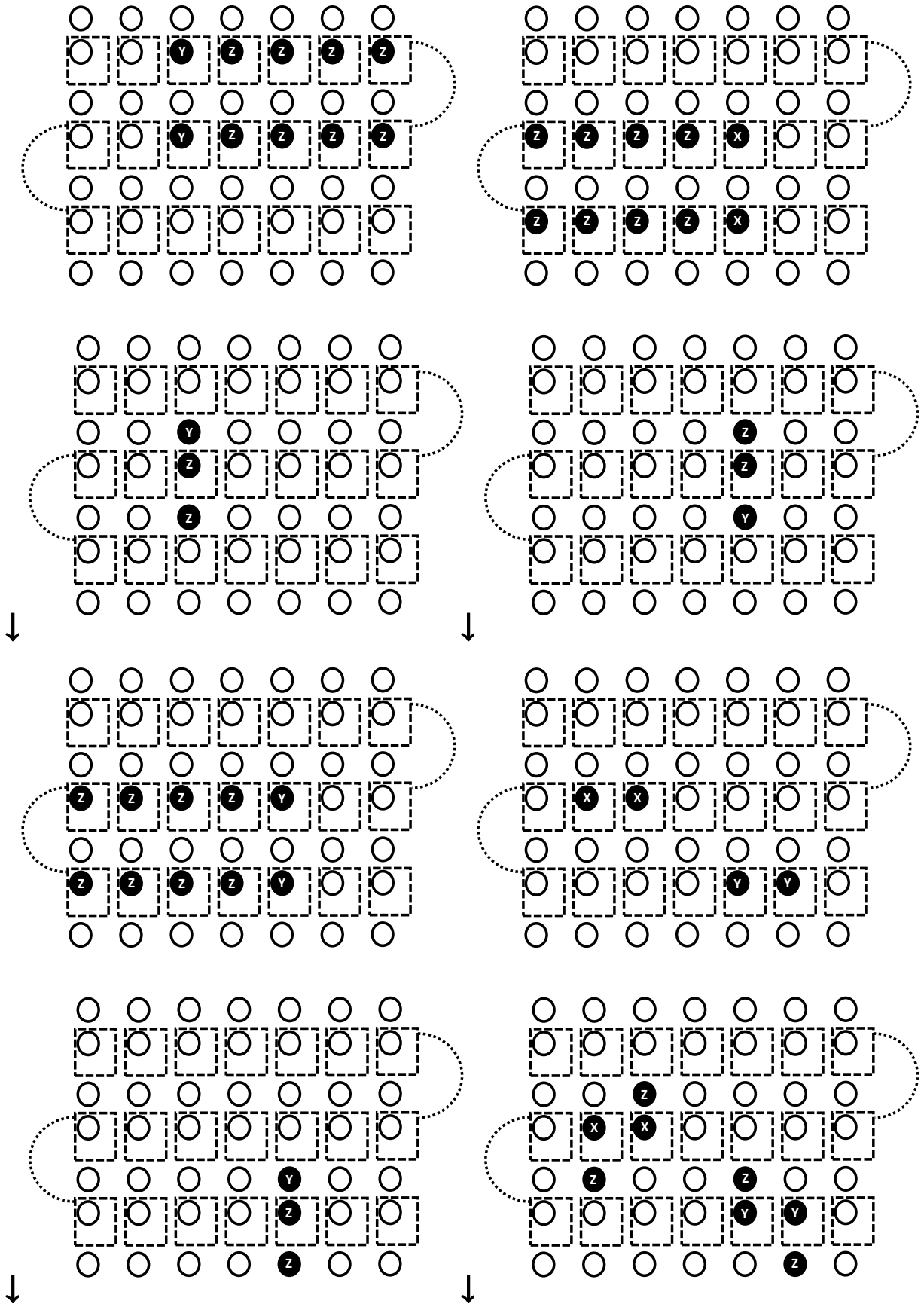


Fig. A.1.4 Diagrams showing how the portion of Pauli-strings acting on the roots of cells are modified and simplified by the Hybrid+ mapping algorithm

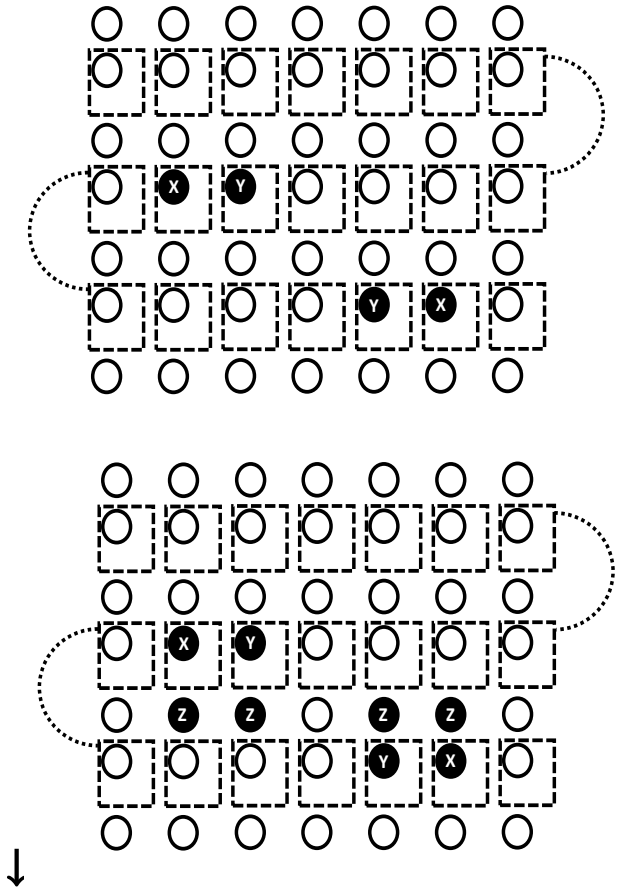


Fig. A.1.5 Diagrams showing how the portion of a Pauli-string acting on the roots of the cells is modified and simplified by the Hybrid+ mapping algorithm

Appendix B

Installing the CUED class file

LaTeX.cls files can be accessed system-wide when they are placed in the <texmf>/tex/latex directory, where <texmf> is the root directory of the user's TeXinstallation. On systems that have a local texmf tree (<texmflocal>), which may be named "texmf-local" or "localtexmf", it may be advisable to install packages in <texmflocal>, rather than <texmf> as the contents of the former, unlike that of the latter, are preserved after the LaTeXsystem is reinstalled and/or upgraded.

It is recommended that the user create a subdirectory <texmf>/tex/latex/CUED for all CUED related LaTeXclass and package files. On some LaTeXsystems, the directory look-up tables will need to be refreshed after making additions or deletions to the system files. For TeXLive systems this is accomplished via executing "texhash" as root. MIKTeXusers can run "initexmf -u" to accomplish the same thing.

Users not willing or able to install the files system-wide can install them in their personal directories, but will then have to provide the path (full or relative) in addition to the filename when referring to them in LaTeX.